Int. J. Electron. Commun. (AEÜ) 66 (2012) 772–779

Contents lists available at SciVerse ScienceDirect

International Journal of Electronics and Communications (AEÜ)

journal homepage: www.elsevier.de/aeue

An efficient method for impulse noise reduction from images using fuzzy cellular automata

Sana Sadeghi [a,*], Alireza Rezvanian [a,b], Ebrahim Kamrani [c]

[a] Department of Electrical Engineering, Faculty of Engineering, Islamic Azad University, Hamedan Branch, Hamedan, Iran
[b] Computer Engineering and Information Technology Department, Amirkabir University of Technology (Tehran Polytechnic), 424 Hafez Ave., Tehran, Iran
[c] Computer Science Department, Dalarna University, 78188 Borlänge, Sweden

ARTICLE INFO

Article history:
Received 4 August 2011
Accepted 15 January 2012

Keywords:
Image processing
Image enhancement
Impulse noise
Impulse noise reduction
Fuzzy cellular automata

ABSTRACT

Impulse noise reduction from corrupted images plays an important role in image processing. This problem will also affect on image segmentation, object detection, edge detection, compression, etc. Generally, median filters or nonlinear filters have been used for noise reduction but these methods will destroy the natural texture and important information in the image like the edges. In this paper, to eliminate impulse noises from noisy images, we used a hybrid method based on cellular automata (CA) and fuzzy logic called Fuzzy Cellular Automata (FCA) in two steps. In the first step, based on statistical information, noisy pixels are detected by CA; then using this information, the noisy pixel will change by FCA. Regularly, CA is used for systems with simple components where the behavior of each component will be defined and updated based on its neighbors. The proposed hybrid method is characterized as simple, robust and parallel which keeps the important details of the image effectively. The proposed approach has been performed on well-known gray scale test images and compared with other conventional and famous algorithms, is more effective.

1. Introduction

Image processing is one of the most important fields of artificial intelligence in the modern world, having many applications in medicine, weather forecasting, machine vision and different industries. One of the important problems with digital images is that impulse noise may occur because of sensors or transfer channels [1]. Noise reduction is an important issue as a prelude to image processing for image segmentation, object recognition, edge detection and compression [2].

There are many methods for reducing impulse noises such as the median filter which is a well-known and widely used conventional method [3]. For noise removal, the median filter replaces the value of the center point by the median value of the neighboring pixels. However, the lack of differentiation between the noisy and non-noisy pixels of the image causes damage to the non-noisy pixels of the image too. Other variations of median filter such as center-weighted median filter [4], switching vector median filter [5], adaptive window length recursive weighted median filter [6] and adaptive switching median filter [7] have been recommended to make a balance between the elimination of noisy pixels and keeping image details. With specific changes, these methods have a better performance compared with the conventional median filter. Although these methods are widely used because of their simplicity, they are effective only where noise possibility is less than 20 percent and have undesirable effects such as the smoothness of important pixels of the image on noise possibilities of more than 20 percent. Considering the specifications of impulse noise and the uncertainty of noise existence on different points of the image, the use of fuzzy logic has an effective role on the noise reduction and image improvement such as [8] which using fuzzy rules tried to replace better value for noisy pixels. One of the most effective methods using fuzzy logic is called fuzzy filter [9] used as the starting point by many researchers. Other different methods using a hybrid of median filter and fuzzy logic in [10–12], fuzzy logic controller in [13] and fuzzy wavelet shrinkage in [14] have been proposed. Some approaches used Adaptive Neuro-Fuzzy Inference System (ANFIS) to remove noise since most of them have two stages [15–17].

Most soft computing methods reduce noises effectively but lack of robustness, high costs of computation, being time-consuming, high complexity, and the need for evaluating and tuning the parameters are some problems of these methods. Cellular Automata is a simple and robust method with parallelization ability used in [18] for image processing applications such as improving image quality. However, despite its simplicity, this method like median filter

* Corresponding author. Tel.: +98 811 429 4145.
E-mail addresses: sana.sadeghi@gmail.com, sadeghi@iauh.ac.ir (S. Sadeghi), a.rezvanian@aut.ac.ir (A. Rezvanian), v08ebrka@du.se (E. Kamrani).

1434-8411/$ – see front matter
doi:10.1016/j.aeue.2012.01.010

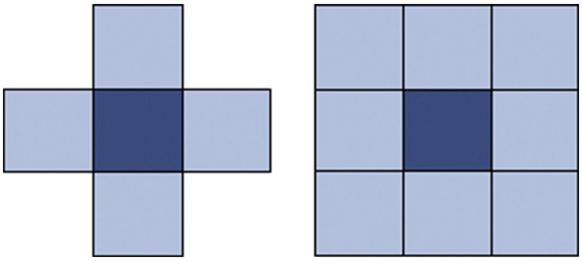

Fig. 1. Neighborhood model: (a) Von Neumann and (b) Moore.

or other noise reduction methods, delivers a good result only when the corrupted noises are low, but in case of a high noise occurrence all effective pixels will change.

In this paper, in the first step, the detection of noisy points will be done using information statistics of neighbor pixels by cellular automata. In the second step, the value of detected noisy pixels will be changed according to their fuzzy membership value of noisy set. In the rest of paper, Cellular Automata (CA) and Fuzzy Cellular Automata (FCA) will be introduced briefly in Sections 2 and 3 respectively. The details of the proposed approach using FCA are presented in Section 4. In Section 5, the experimental results of the proposed method on well known gray scale test images are illustrated and its performance is compared with other popular and conventional methods.

2. Cellular automata

In the late 1940, cellular automata (CA), was described by Von Neumann and Olman presented cellular automata as a model for investigating the behavior of complex systems. A cellular automaton is a mathematical model for systems with multiple simple components which collaborate to make complex patterns. The behavior of each component is defined based on the behavior its neighbors. The cellular automaton is a discrete, decentralized, self-organized system which can make an ordered structure starting on a stochastic state. This system can reduce entropy during time. It should be mentioned that in the cellular automata, computations are done in a parallel manner [19]. In this model, space is defined as a grid of cells in which each cell is a state memory. In each step, the cell considers the adjacent neighbors and using transmission state rules takes its next discrete state. Besides every cell can process independently from the other cells.

There are different neighborhood structures for cellular automata. In general, we can consider each ordered set of cells as neighbors. The first neighborhood type is Von Neumann which includes 4 non-angular cells related to central cell. The other neighborhood type is Moore which includes all 8 neighbors around the cell. These two types of neighborhoods are the most famous ones called nearest neighbors shown in Fig. 1.

In the following section, the proposed method using hybridization of cellular automata and fuzzy logic is presented.

Following is the proposed method using hybridization of cellular automata and fuzzy logic.

3. Fuzzy cellular automata

Considering the ability and usage of fuzzy logic in processing uncertain data, a fuzzy cellular automata (FCA) structure is introduced in which instead of using certain values in the cells and their transition functions, uncertain and fuzzy values are used. In other words, FCA is a CA in which fuzzy logic is applied to the states of the cells and transmission state rules [20]. There are different definitions for fuzzy cellular automata [21,22]. In this paper, the latest definition is used [22]. In FCA all states of a cell and local transition function (rules) are fuzzy. In fact, all states of a cell are linguistic variables defined based on our knowledge of problem domain. The next state of each cell depends on the current state of itself and its neighbors.

This change of states is due to the local transition function which is a fuzzy function and the same for all the cells. This function uses the current membership value of each neighbor to calculate it for the next step. To demonstrate the evolution progress of FCA, the membership value of the linguistic variables of the cell in each moment is used. The neighborhood is the same for all the cells and remains unchanged in time.

FCA can be defined as a quaternary of $\{Z, S, r, f\}$ where Z is a regular grid of cells with n dimension, S is the state set of each cell in [0, 1] range ($r...N$) is neighborhood radius and f: s^{2r+1}:→ s is the fuzzy transition function.

4. Proposed method

Digital image processing in many applications is considered a real-time process in which the process speed is very important. Therefore, the parallel algorithms in image processing are much more important compared with serial algorithms. In [23], a CA is used as a parallel method for image edge detection. An application of cellular automata as a fundamental method for image enhancement is introduced in [24]. CA is used as a method for noise removal in digital images [18]. Although these methods are simple because of using median to replace the value of noisy places, they are useful only when noise density is very low and they do not render a good performance when the noise density is medium or high.

One of the main problems in most noise elimination methods is the elimination of important data and details of the image such as the edges and zone texture. In addition, in conventional methods such as median filter and its variations, the image will take damage because without any information about the noisy points, the correct pixels are also changed. Considering the uncertainty and ambiguity of impulse noise in digital images, fuzzy logic can have a suitable behavior in such situations. In the proposed method, FCA which had never been used before in noise removal applications is used. The proposed method works in two steps; at first, the noisy pixels are detected by CA (discussed in Section 4.1); in the second step, the detected noisy pixels are replaced by FCA (presented in Section 4.2). The flow chart of the proposed method is shown in Fig. 2.

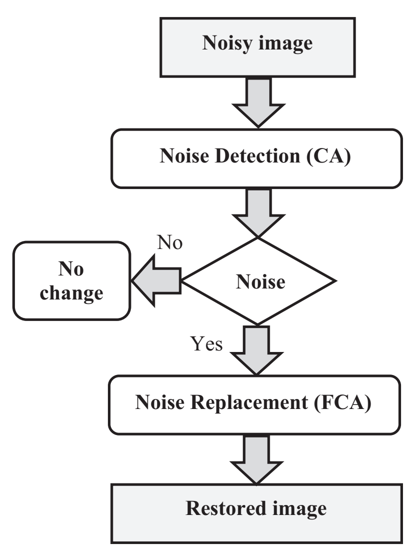

Fig. 2. The flow chart of proposed method.

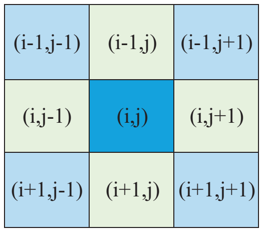

Fig. 3. Central cell parametric position and the neighbors.

4.1. *Noise detection using cellular automata*

In the most approaches in noise removal, the noises were detected by a supervised process, which one phase allocated for training the noise pixels. In these methods, parameter tuning, good training, and computational challenges are appeared. The proposed approach using cellular automata provide parallel framework computation in each cell without any preprocessing and learning phase.

One of the main specifications of impulse noise is that each noisy point has a significant change compared with its neighbors. Therefore, in the preprocessing section, noise density is obtained using statistical information such as average and standard deviation of central and neighbor cells. The equation is:

$$D_i = \left|X_c - X_i\right| \; 1 \leq i \leq 8 \tag{1}$$

where X_i includes neighborhood pixels that are shown in Fig. 3 and Xc is the value of the central cell. D_i is the distance value for neighboring pixels from the central cell. Using different neighborhood types results in different neighborhood pixels.

Average and standard deviation which are two important statistical factors are calculated using the following equations:

$$N_m = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{2}$$

$$N_s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(D_i - \bar{D})^2} \tag{3}$$

In Eqs. (2) and (3), N_m is the average of neighborhood values of the central cell and N_s is the standard deviation of the neighboring points from the central cell. D_i is calculated in Eq. (1) and $\bar{D}$ is the average value of D_i. Also, the value of n can be 4 or 8 according to different choices for neighborhood type. The existence of a high distribution related to average value can be estimated according to a threshold value. Here, we used two-dimensional cellular automata including a square network with Von Neumann neighborhood structure for detecting noisy pixels. The cell states

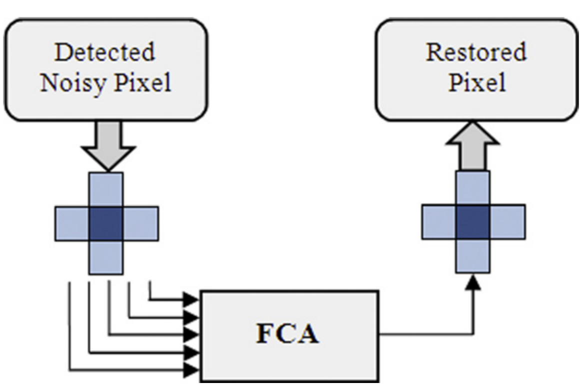

Fig. 4. General structure of FCA method using fuzzy cellular automata.

Table 1
Comparison of the PSNR values for restoration results of the various methods for the *Lena* test image.

Methods	Noise ratio							
	10%	20%	30%	40%	50%	60%	70%	80%
Noisy	15.5	12.4	10.6	9.5	8.5	7.7	7.0	6.4
SMF3	33.3	29.0	23.4	19.1	15.4	12.3	10.1	8.1
SMF5	30.5	28.9	27.2	25.7	23.0	18.4	14.2	10.3
CWMF	29.3	24.0	19.5	15.8	13.0	10.8	8.9	7.6
IRF	30.3	27.1	22.5	18.2	14.9	12.2	9.8	8.1
SWMF	32.9	28.3	23.4	18.6	15.3	12.6	10.1	8.4
SDROMF	30.6	28.5	25.8	23.5	20.7	17.8	14.2	10.6
PSMF	30.7	28.8	26.9	23.8	20.0	15.3	11.1	8.4
FF	31.0	27.6	24.9	22.7	20.3	17.7	14.8	11.7
FIRE	29.9	23.2	18.8	15.5	12.8	10.9	9.3	7.9
IFCF	32.6	29.0	25.0	21.3	18.1	15.6	13.5	11.7
FSF	33.7	28.9	23.0	18.1	14.3	11.5	9.3	7.4
SNFF	30.1	28.4	24.0	21.9	20.0	18.3	16.7	15.2
NNANFIS	34.9	31.5	29.1	27.5	25.4	23.2	20.9	18.0
ANFISFWS	29.2	28.2	26.8	23.8	22.0	15.4	12.9	10.8
NNDMF	32.3	32.0	31.5	30.0	28.9	27.7	26.1	23.6
NFDMF	43.2	39.8	37.6	35.8	34.0	32.3	30.3	27.6
TCA	32.1	30.5	27.7	25.1	20.1	19.5	16.7	15.0
FCA	35.0	33.5	32.7	31.2	29.2	27.8	26.5	23.7

Table 2
Comparison of the PSNR values for restoration results of the various methods for the *Peppers* test image.

Methods	Noise ratio							
	10%	20%	30%	40%	50%	60%	70%	80%
Noisy	15.3	12.3	10.5	9.3	8.3	7.5	6.8	6.3
SMF3	32.8	28.8	23.2	18.7	15.1	12.2	9.8	8.0
SMF5	31.1	29.2	27.7	25.7	22.7	18.5	14.1	10.2
CWMF	28.9	23.3	19.0	15.4	12.6	10.5	8.6	7.7
IRF	30.3	25.5	21.7	17.6	14.5	11.8	9.4	7.8
SWMF	32.3	27.4	22.8	18.3	15.1	12.2	9.7	8.0
SDROMF	30.3	26.4	23.8	21.4	19.4	16.2	13.1	10.2
PSMF	30.7	28.4	26.1	23.3	19.6	15.2	10.9	8.3
FF	29.6	26.1	23.4	21.1	19.0	16.6	14.1	11.3
FIRE	29.7	25.6	20.9	17.0	12.5	10.7	9.4	8.0
IFCF	26.9	22.3	19.5	18.7	17.2	15.5	11.5	10.7
FSF	27.4	25.8	24.6	18.5	17.0	15.3	11.9	9.2
SNFF	24.5	23.1	21.4	19.9	18.4	16.8	15.5	14.2
NNANFIS	27.3	26.2	24.9	22.7	19.5	17.6	15.9	13.6
ANFISFWS	28.7	27.3	26.8	25.5	23.2	17.7	14.4	12.1
NNDMF	25.6	24.6	23.4	22.6	21.8	21.1	19.4	18.1
NFMDF	33.5	31.8	30.4	29.6	27.8	26.0	22.3	20.7
TCA	31.1	29.3	26.9	24.6	22.0	19.3	16.2	13.2
FCA	32.0	31.5	29.9	29.1	27.2	24.6	20.1	18.2

Table 3
Comparison of the PSNR values for restoration results of the various methods for the *Baboon* test image.

Methods	Noise Ratio							
	10%	20%	30%	40%	50%	60%	70%	80%
Noisy	15.6	12.6	10.8	9.5	8.6	7.8	7.1	6.5
SMF3	23.1	22.0	20.1	17.3	14.5	12.1	9.9	8.2
SMF5	21.2	20.9	20.4	19.9	18.8	16.5	13.3	10.1
CWMF	25.5	22.7	18.9	15.6	12.9	10.9	9.1	7.7
IRF	25.8	24.1	21.2	17.8	14.7	12.3	9.9	8.2
SWMF	28.8	26.0	22.1	18.5	15.2	12.7	10.2	8.4
SDROMF	25.8	24.6	23.2	21.8	19.7	17.6	14.2	10.8
PSMF	27.8	26.4	25.0	22.9	19.5	15.3	11.0	8.3
FF	29.5	26.7	24.3	22.4	20.1	17.7	15.0	11.9
FIRE	24.9	21.4	18.1	15.3	13.1	11.3	9.8	8.6
IFCF	23.5	22.0	20.1	18.0	15.8	14.1	12.7	11.8
FSF	23.0	22.1	20.0	17.1	14.3	11.9	9.8	8.0
SNFF	30.9	28.2	26.2	24.3	22.4	20.8	19.2	17.8
NNANFIS	31.1	28.0	26.1	24.5	23.0	21.5	19.6	17.2
ANFISFWS	28.7	23.8	23.0	22.1	19.9	18.1	15.4	13.9
NNDMF	22.7	22.6	22.5	22.3	22.0	21.6	21.2	20.5
NFDMF	32.3	29.3	27.4	26.0	24.7	23.6	22.5	21.3
TCA	29.1	26.0	24.8	22.3	18.5	15.6	11.5	10.1
FCA	31.2	28.5	27.2	24.6	23.3	22.3	21.5	21.0

Fig. 5. Comparison of the Restored images for the *Lena* by impulse noise density of 50%.

Table 4
Comparison of the *MSE* values for restoration results of the cellular based methods for the *Bridge* test image.

Methods	Noise ratio							
	10%	20%	30%	40%	50%	60%	70%	80%
Noisy	1956.64	3915.67	5858.61	7761.63	9679.75	11658.47	13533.63	15568.21
SMF3	274.44	309.76	361.16	439.37	621.86	1285.78	2829.37	6528.97
TCAV	226.39	256.18	318.85	433.79	625.62	1029.54	1754.27	3444.13
TCAM	188.18	240.06	343.53	541.72	855.98	1483.83	2465.55	4558.50
FCAV	180.79	190.54	206.52	232.90	286.43	430.20	804.56	1980.84
FCAM	207.57	219.17	234.66	258.76	306.88	423.34	740.41	1758.92

Table 5
Comparison of the *MSE* values for restoration results of the cellular based methods for the *Boat* test image.

Methods	Noise ratio							
	10%	20%	30%	40%	50%	60%	70%	80%
Noisy	1836.26	3704.53	5546.76	7356.94	9207.36	11087.92	12952.81	14739.48
SMF3	136.44	167.22	213.29	289.49	465.43	1085.49	2742.68	6076.52
TCAV	125.25	149.19	231.26	391.19	690.85	1242.51	2191.45	4054.23
TCAM	107.89	146.77	195.78	286.51	455.76	811.86	1478.38	3026.62
FCAV	81.64	86.53	96.41	113.21	145.74	257.14	551.90	1583.00
FCAM	98.39	104.41	114.31	130.08	159.31	257.79	517.19	1413.40

Fig. 6. Comparison of the Restored images for the *Peppers* by impulse noise density of 50%.

are updated in discrete time steps based on transition function F which is related to the current states of the neighbors. Eq. (4) shows the operation function based on time and neighborhood pixels.

$$S_{i,j}^{t+1} = F[S_{i,j-1}^{t}, S_{i,j+1}^{t}, S_{i-1,j}^{t}, S_{i+1,j}^{t}, S_{i-1,j-1}^{t}, S_{i-1,j+1}^{t}, S_{i+1,j-1}^{t}, S_{i+1,j+1}^{t}] \quad (4)$$

In this relation $S_{i,j}^{t+1}$ is the next state of cell in i,j coordination. In Fig. 3, the central cell parametric position and its neighbors are presented.

The result value of the function shows the distribution of pixels which is more than a threshold value in a noisy state. Eq. (5) shows how this function is obtained.

$$F(i,j) = \frac{N_m}{N_s} \quad (5)$$

Finally, the ratio of statistical parameters is in $F(i,j)$. Eq. (6) shows how the noisy points are obtained.

$$\text{Nosie} \quad (i,j) = \begin{cases} 1 & F(i,j) \geq \theta \\ 0 & F(i,j) < \theta \end{cases} \quad (6)$$

where Noise(i,j) demonstrates the existence of noise in i, j coordination. $F(i,j)$ is the ratio of statistical parameters resulted from Eq. (5) and θ is the threshold value for noise existence. Therefore, by assuming a threshold (obtained from the experiments), the noisy points can be approximately detected.

4.2. Noise estimation using fuzzy cellular automata

A more logical and suitable behavior is obtained with fuzzy values for image points in a neighborhood of cellular automata. In the proposed method, a two dimensional fuzzy cellular automata is used including the square network of Von Neumann neighborhood structure. The cells' state is updated in discrete time steps based on transition function F which depends on the current states of the neighbors. According to Weber's law, human's eye cannot distinguish the details of image in an identical light background [1]. Therefore, the operational function for noisy value replacement is based on time and coordination of neighborhood points as in Eq. (7).

$$S_{i,j}^{t+1} = F[S_{i,j-1}^{t}, S_{i,j+1}^{t}, S_{i,j-1}^{t}, S_{i+1,j}^{t}] \quad (7)$$

The general structure of the FCA method is presented in Fig. 4.

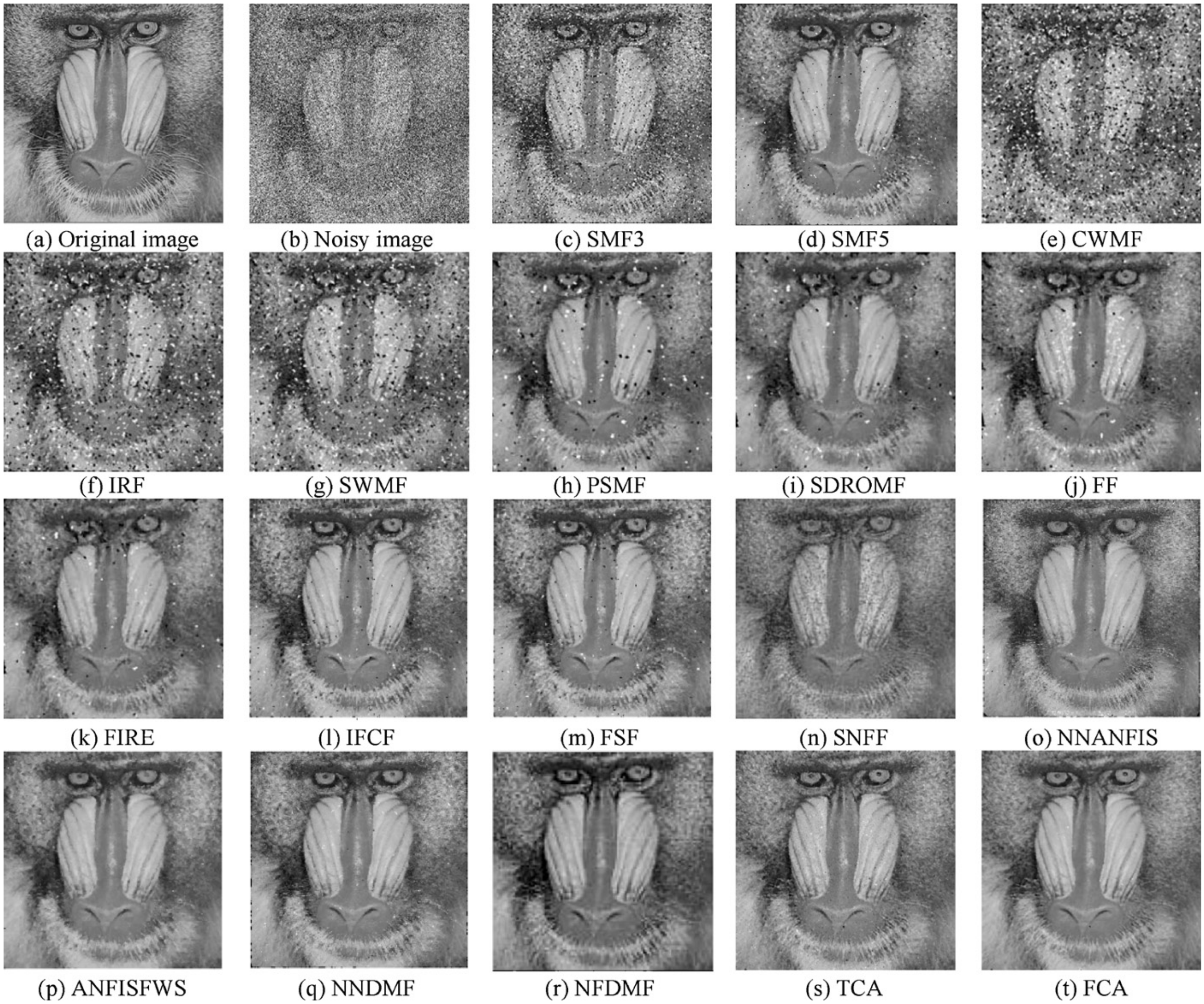

Fig. 7. Comparison of the Restored images for the *Baboon* by impulse noise with 50% probability of noise.

Therefore, Eq. (8) is used for fuzzifying the points' membership value in relation to noisy set based on Von Neumann neighborhood structure:

$$M_i(x) = \frac{|w - x_i|}{w}, \quad 0 \le x \le 255 \tag{8}$$

where $M_i(x)$ is the value of fuzzy membership function, X_i is value of neighborhood points, and the value of W, according to Weber's law, is assumed 127.5 here.

Eq. (9) is used for defuzzifying and replacement value based on the average center of gravity (9).

$$Y = \frac{\sum_{i=1}^{n} M_i \cdot x_i}{\sum_{i=1}^{n} M_i}, \quad 0 \le x \le 255 \tag{9}$$

where Y is the new value of the pixel. According to fuzzy membership values and correlation principle in correct neighborhood pixels, the return value of defuzzyfying function has been defined. Assuming a threshold value for convergence in changes of pixels, obtained from the experiments, we can repeat the proposed method to obtain a stable state.

The results of the proposed method, in comparison with those of other conventional methods, are presented in the next section.

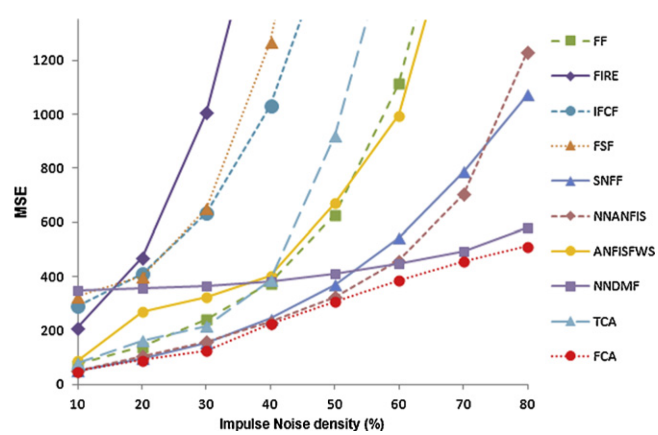

Fig. 8. Performance comparison of selective methods and proposed method as a function of noise ratio for *Lena* image.

5. Experimental results

The structure and details of the proposed method using FCA were described in the previous section. In this section, the results of applying the proposed method to famous gray level test images, *Lena*, *Peppers* and *Baboon* are investigated.

Performance of proposed method compared with other conventional and state of the art methods such as *Standard Median Filter* with 3×3 mask and 5×5 mask as (SMF3) and (SMF5) respectively [25], *Center Weighted Median Filter* (CWMF) [26], *Impulse Rejecting Filter* (IRF) [27], *Switching Median Filter* (SWMF) [28], *Signal Dependent Rank Order Mean Filter* (SDROMF) [29], *Progressive Switching Median Filter* (PSMF) [30], *Fuzzy Filter* (FF) [9], *Fuzzy Inference Ruled by Else-action* (FIRE) [31], *Iterative Fuzzy Control based Filter* (IFCF) [32], *Fuzzy Similarity Filter* (FSF) [33], *Simple Neuro-Fuzzy Filter* (SNFF) [18], *Neural Network Detector Median Filter* (NNDMF) [17], *Neuro-Fuzzy Detector Median Filter* (NFDMF) [17], *Neural Network Detector Adaptive Neuro-Fuzzy Inference System* (NNANFIS) [16], *Adaptive Neuro-Fuzzy System and Fuzzy Wavelet Shrinkage* (ANFIS-FWS) [14] *Traditional Cellular Automata* with noise detection (TCA) [34] and proposed method as FCA. Subjective quantitative measure as visual are presented by restored noisy images with probability of 50% in Fig. 5 for *Lena*, in Fig. 6 for *Peppers* and in Fig. 7 for *Baboon*.

In addition to visual criteria, the performance of the proposed method in comparison with the other methods is evaluated by *Peak Signal-to-Noise Ratio* (PSNR) which is calculated by Eq. (10).

$$\mathrm{PSNR} = 10 \quad \log_{10}\left(\frac{255^2}{\mathrm{MSE}}\right) \tag{10}$$

Mean Squared Error (MSE) as an objective quantitative criterion is presented as follows (11):

$$\mathrm{MSE} = \frac{\sum_{i=1}^{N}\sum_{j=1}^{M}(X(i,j) - Y(i,j))^2}{M \times N} \tag{11}$$

where $X(i,j)$ is the pixel of the original noisy image, $Y(i,j)$ is the pixel of the restored noisy image in i, j coordination. N and M are the image size. The resulted *Mean Squared Error* (MSE) values for *Lena*, *Peppers* and *Baboon* are listed in Tables 1–3 respectively for various states of noise density from 10% to 80%.

According to visual images and numerical values, FCA has better performances in comparison with the other popular and state-of-the-art methods except NFDMF.

Also, Fig. 8 presents another comparison with some other methods demonstrating that the proposed method with noise rate growth based on PSNR shows the suitability and noise changing rate very well.

Another experiments accomplished on the cellular based methods. In this experiments proposed method compared with different neighbors, so the performance of Traditional Cellular Automata with Von Neumann as TCAV, Traditional Cellular Automata with Moore as TCAM, Fuzzy Cellular Automata with Moor as FCAM, Fuzzy Cellular Automata with Von Neumann as FCAV, are reported by MSE for Bridge and Boat in Tables 4 and 5 respectively.

Although there is association with fuzzification type, neighborhood structure as in baboon image maybe need small neighborhood and more fuzzy rules, As in experiments, proper results achieved by proposed method with current setting of method. The focus of proposed method is considered simplicity and parallel manner of method by cellular automata.

6. Conclusion

In this paper, a hybrid method using cellular automata and fuzzy logic is proposed. Simplicity, robustness, parallel manner and distribution ability for noise enhancement using fuzzy cellular automata are the main advantages of the proposed method. In previous conventional methods, without considering boundary pixels, changes usually occurred in the central value related to neighbors but, in the proposed method noise existence affects the central value after considering its neighbors using fuzzy cellular automata resulting in a suitable performance. The proposed method in this paper using Von Neumann neighborhood structure in fuzzy cellular automata was applied to famous gray level test images. Visual criteria and numerical criteria of the proposed method were compared with those of other famous methods and most results demonstrate improvements for the proposed method. Using other different neighborhood structures and hybridizing these methods for eliminating specific problems in high noisy images are the concerns of the authors for more future research.

Acknowledgements

The authors thank Mr. Arash Moradi for some improvements and reviewers for usefull comments.

References

[1] Dharmarajan R, Kannan K. A hypergraph-based algorithm for image restoration from salt and pepper noise. Int J Electron Commun (AEU) 2010;64:1114–22.
[2] Tavassoli S, Yaqubi M, Rezvanian A, Ebadzadeh MM. Enhancement of ear verification system performance using a new hybrid denoising approach (ANFIS-FWS). In: Proceedings of IEEE 2010 1st International Conference on Integrated Intelligent Computing. 2010. p. 200–4.
[3] Zhang J. An efficient median filter based method for removing random-valued impulse noise. Digital Signal Process 2010;20:1010–8.
[4] Yin L, Yang R, Gabbouj M, Neuvo Y. Weighted median filters: a tutorial. IEEE Trans Circuits Syst II: Anal Digit Signal Process 1996;43:157–92.
[5] Celebi ME, Aslandogan YA. Robust switching vector median filter for impulsive noise removal. J Electronic Imaging 2008;17:043006–43011.
[6] Kumar VRV, Manikandan S, Vanathi PT. Adaptive window length recursive weighted median filter for removing impulse noise in images with details preservation. Ecti Trans Elec Eng Electron Commun 2008;6:73–80.
[7] Akkoul S, Ledee R, Leconge R, Harba R. A new adaptive switching median filter. IEEE Signal Process Lett 2010;17:587–90.
[8] Schulte S, Nachtegael M, De Witte V, Van der Weken D, Kerre EE. A fuzzy impulse noise detection and reduction method. IEEE Trans Image Process 2006;15:1153–62.
[9] Russo F, Ramponi G. A fuzzy filter for images corrupted by impulse noise. IEEE Signal Process Lett 1996;3:168–70.
[10] Kang CC, Wang WJ. Fuzzy reasoning-based directional median filter design. Signal Process 2009;89:344–51.
[11] Toh K, Isa N. Noise adaptive fuzzy switching median filter for salt-and-pepper noise reduction. Signal Process Lett IEEE 2010;17:281–4.
[12] Toh K, Ibrahim H, Mahyuddin MN. Salt-and-pepper noise detection and reduction using fuzzy switching median filter. IEEE Trans Cons Electronics 2008;54:1956–61.
[13] Hussain A, Masood Bhatti S, Jaffar MA. Fuzzy based impulse noise reduction method. Multimedia Tools Appl 2012:1–21, doi:10.1007/s11042.011.0829.7.
[14] Tavassoli S, Rezvanian A, Ebadzadeh MM. A new method for impulse noise reduction from digital images based on adaptive neuro-fuzzy system and fuzzy wavelet shrinkage. In: Proceedings of IEEE 2nd 2010 International Conference on Computer Engineering and Technology, vol. 4. 2010. p. 297–301.
[15] Wang X, Zhao X, Guo F, Ma J. Impulsive noise detection by double noise detector and removal using adaptive neural-fuzzy inference system. Int J Electron Commun (AEU) 2010;65:429–39.
[16] Rezvanian A, Faez K, Mahmoudi F. A two-pass method to impulse noise reduction from digital images based on neural networks. In: Proceedings of IEEE International Conference on Electrical and Computer Engineering. 2008. p. 400–5.
[17] Turkmen I. Efficient impulse noise detection method with ANFIS for accurate image restoration. Int J Electron Commun (AEU) 2010;65:132–9.
[18] Rosin PL. Image processing using 3-state cellular automata. Comp Vision Image Unders 2010;114:790–802.
[19] Wolfram S. Theory and applications of cellular automata 1986.
[20] Enayatifar R, Meybodi MR. Adaptive edge detection via image statistic features and hybrid model of fuzzy cellular automata and cellular learning automata. In: 2009 International Conference on Information and Multimedia Technology. 2009. p. 273–8.
[21] Pradipta M, Chaudhuri PP. Fuzzy cellular automata for modeling pattern classifier. IEICE Trans Inf Syst 2005;88:691.
[22] Anvarinejhad T, Meybodi MR. Fuzzy cellular automata. In: Proceedings of the 5th Iranian Conference on Fuzzy Systems. 2004. p. 57–65.
[23] Chang C, Zhang Y, Gdong YY. Cellular automata for edge detection of images. In: Machine Learning and Cybernetics, 2004. Proceedings of 2004 International Conference, vol. 6. 2004. p. 3830–4.
[24] Hernandez G, Herrmann HJ. Cellular automata for elementary image enhancement. Graphic Models Image Process 1996;58:82–9.

[25] Gabbouj M, Coyle EJ, Gallagher NC. An overview of median and stack filtering. Circuit Syst Signal Process 1992;11:7–45.
[26] Ko SJ, Lee YH. Center weighted median filters and their applications to image enhancement. IEEE Trans Circuits Syst 1991;38:984–93.
[27] Chen T, Wu HR. Adaptive impulse detection using center-weighted median filters. IEEE Signal Process Lett 2001;8:1–3.
[28] Zhang S, Karim MA. A new impulse detector for switching median filters. IEEE Signal Process Lett 2002;9:360–3.
[29] Abreu E. Signal-dependent rank-ordered mean (SD-ROM) filter. Nonlinear Image Process 2001:111–34.
[30] Wang Z, Zhang D. Progressive switching median filter for the removal of impulse noise from highly corrupted images. IEEE Trans Circuits Syst II: Anal Dig Signal Process 1999;46:78–80.
[31] Russo F. FIRE operators for image processing. Fuzzy Sets Syst 1999;103: 265–75.
[32] Farbiz F, Menhaj MB. A Fuzzy Logic Control based Approach for image Filtering. Fuzzy techniques in image processing. Springer; 2000. p. 194–221.
[33] Kalaykov I, Tolt G. Real-time image noise cancellation based on fuzzy similarity. Fuzzy filters for image processing. Springer; 2003. p. 54–71.
[34] Rezvanian A, Rezvanian S, Khotanlou H. A new method to impulse noise reduction from medical images using cellular automata. In: Proceedings of the 17th Iranian Conference on Electrical Engineering, ICEE, vol. 8. 2009. p. 53–8.

Sana Sadeghi is a lecturer in the Department of Electrical Engineering, Islamic Azad University of Hamedan. He received the B.Sc. degree in Electrical Engineering (Power Engineering) from Bu-Ali Sina University of Hamedan, Iran in 2007 and the M.Sc. degree from the Shahrood University of Technology 2010, Shahrood Iran. His research interests include distributed generation, distribution system, intelligent systems, and signal processing.

Alireza Rezvanian is obtained his Bachelor's degree in Computer Engineering (Software Engineering) from Bu-Ali Sina University of Hamedan, Iran in 2007 and his masters in Computer Engineering (Artificial Intelligence) from Islamic Azad University of Qazvin in 2010. Currently he is a Ph.D. student in Computer Engineering (Artificial Intelligence) at the Computer Engineering & Information Technology Department, Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran. His research activities include soft computing, evolutionary computation, learning systems and image processing.

Ebrahim Kamrani received the B.Sc. degree in Computer Engineering (Software Engineering) from Islamic Azad University, Tehran Central branch, Iran in 2007 and the M.Sc. degree in Computer Science (Applied Artificial Intelligence) from Dalarna University, Sweden, in 2010. His research interests include soft computing, evolutionary algorithms, fuzzy systems, optimization and Signal Processing.